Group 8 project

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1 Introduction

This investigation uses data from the 2008 - 2012 Scottish Health Survey, which collects information on the health, socio-economic, and lifestyle factors of people living in Scotland. The factors included in this dataset are: age group, sex, employment status, meets recommended vegetable intake (Yes/No), meets recommended fruit intake (Yes/No), survey year, and body mass index (BMI).

Body mass index is a quantity derived from the mass and height of a person, calculated as mass divided by the squared height, and measured in kg/m^2. It is a useful indicator for determining if an individual is underweight (BMI under 18.5), overweight (BMI over 25), or at a healthy weight, and it is especially useful for health data.

The aim of this investigation is to determine whether the body mass index in Scotland has changed between 2008 and 2012, and to assess if there are any differences in the BMI distribution by age, gender, socio-economic factors (employment status) or lifestyle factors (meets recommended fruit and veg intake).

2 Exploratory Analysis

2.1 Data Summaries

We first inspect the balance of the data and check for missing values. The dataset contains a total of 25224 participants. There were no missing values in any of the variables used in this report, so all 25224 participants were retained for both the exploratory analysis and the subsequent modelling.

Table 1 displays the summary statistics for BMI. The mean and median are very similar (27.81 and 27.17 respectively), indicating that the distribution is approximately symmetric, while the standard deviation of 5.26 suggests a large variation in BMI across individuals. The maximum and minimum values of 12.59 and 59.11 respectively indicate a wide range of BMI values within the sample.

Table 1: Summary statistics for BMI

Mean	SD	Median	Min	Max
27.811	5.257	27.173	12.593	59.112

2.2 Distribution Of BMI

Figure 1 shows BMI's distribution is slightly right-skewed and peaks around 26-28. We can observe that the majority of values fall between 22-32, with a large portion of individuals being classifiable as overweight. Highlighted by the peak falling within the overweight grouping.

2.3 BMI and Year

Figure 2 shows the distribution of BMI across the different years of the Scottish Health Survey. The median BMI appears to remain stable over the entire period with no clear trend. The interquartile ranges and overall spread of the data are similar across the entire period. There is little evidence to suggest that BMI changed over the study period, but formal analysis is required to confirm this.

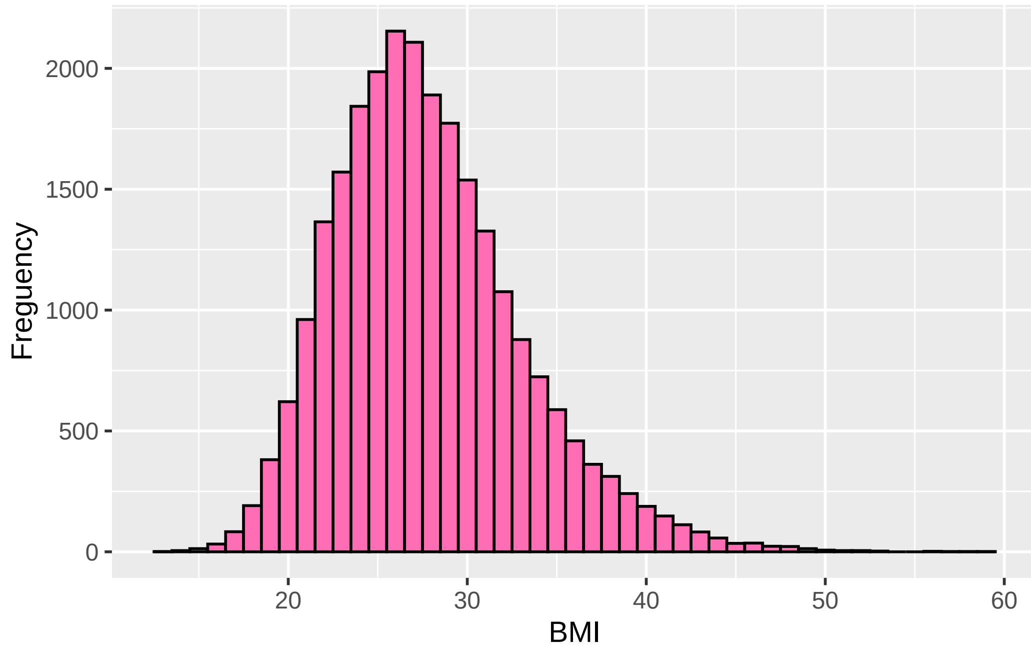

Figure 1: Histogram showing distribution of BMI.

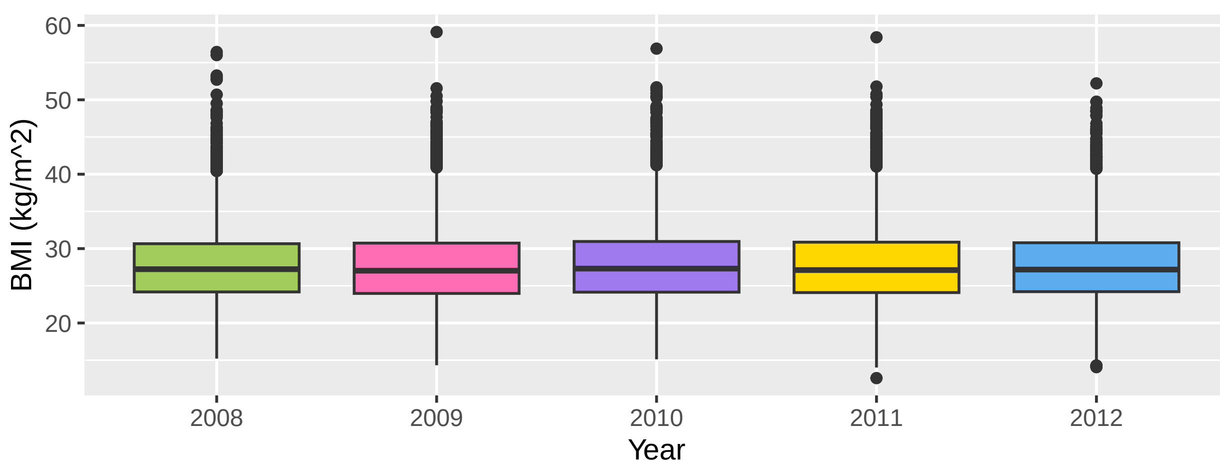

Figure 2: Boxplots of BMI for each year of survey

2.4 BMI and Additional Explanatory Variables

2.4.1 Demographic Variables

Figure 3 shows the distribution of BMI by age group and sex. There appears to be a very slight increase in the median BMI as age group increases, suggesting that older individuals tend to have higher BMI values. But the interquartile ranges have a significant overlap so the differences between age groups may be small or insignificant and should further investigated in later formal analysis.

There appears to be little difference in the distribution of BMI for female and male participants, other than in the interquartile ranges, with a larger range in the female participants, indicating greater variability in BMI among females compared to males.

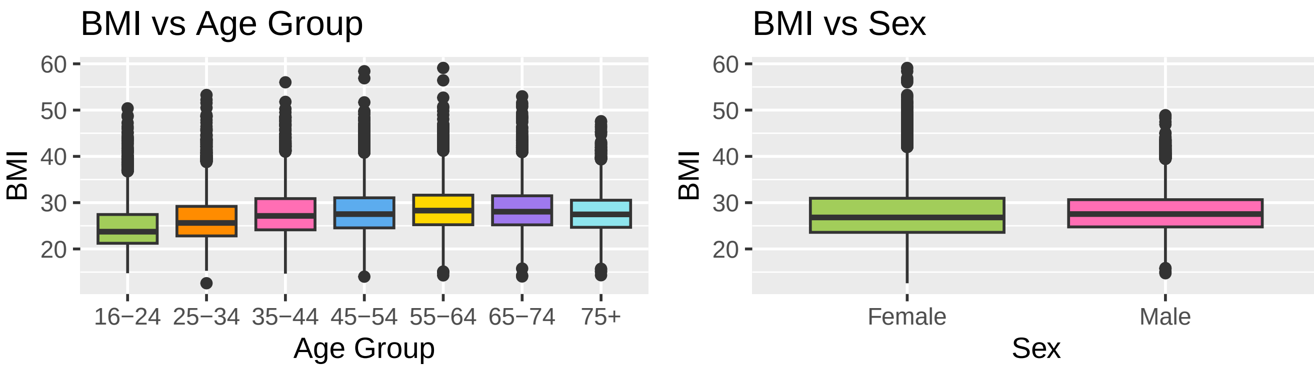

Figure 3: Boxplots of BMI across the demographic variables, age group and sex

2.4.2 Socio-Economic and Lifestyle Variables

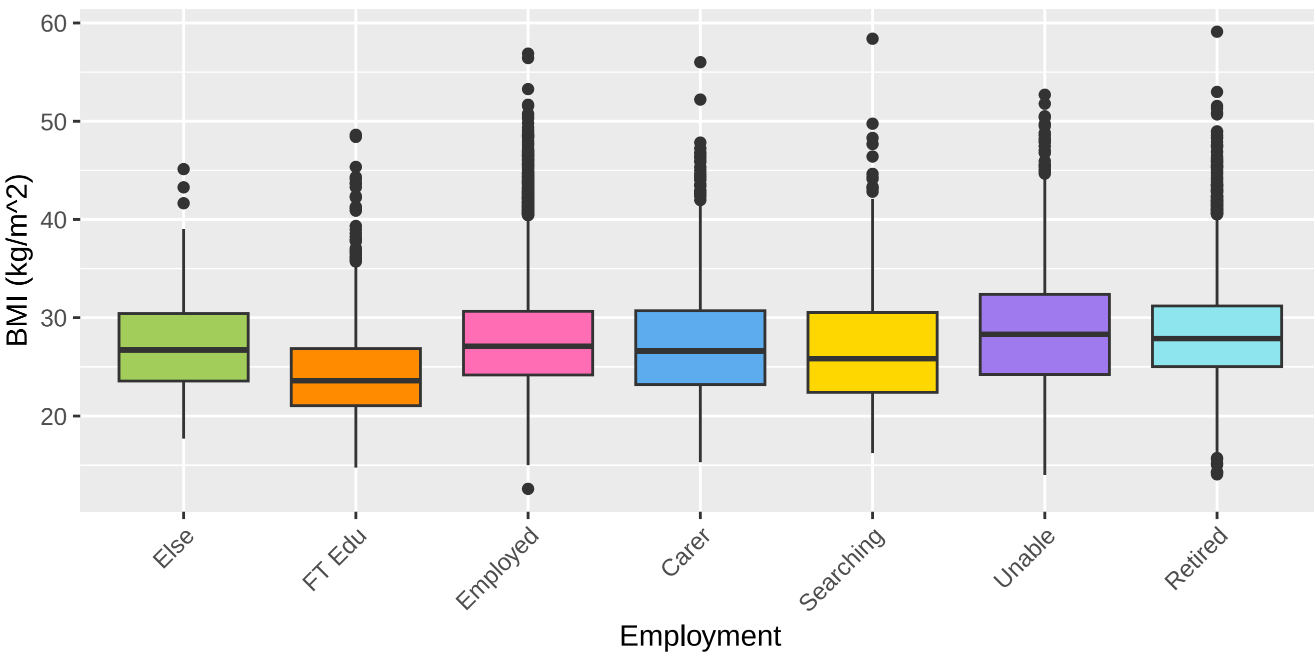

Figure 4: Boxplots of BMI across the socio-economic variable, employment

Figure 4 shows the distribution of BMI across the different employment categories. Individuals in full-time education appear to have a lower BMI compared to other employment groups, but the interquartile ranges for all groups substantially overlap, suggesting that the differences in BMI across these groups are relatively small and significance should be investigated with formal analysis. There also appear to be outliers at both extremes of BMI across the categories.

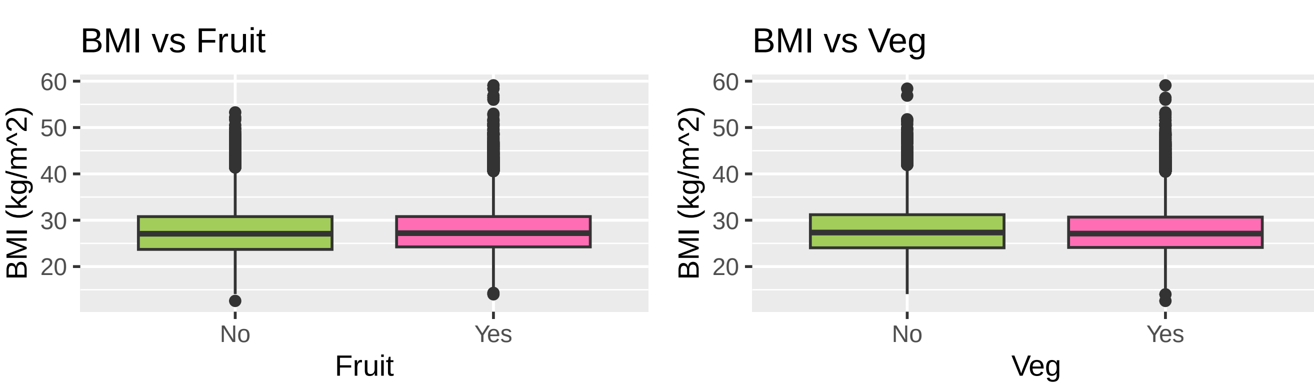

Figure 5: Boxplots of BMI across the lifestyle variables fruit and veg

Figure 5 shows the distribution of BMI by fruit and vegetable consumption. The boxplots in both factors appear very similar with the medians, and interquartile ranges being at similar levels. This suggests that there is no clear evidence of a difference in BMI for participants who consume their recommemded fruit and veg and those who don't, but this should be investigated in formal analysis.

3 Formal Data Analysis

3.1 Full Multiple Regression Model

A full multiple regression model was fitted with BMI as the response variable and all explanatory variables as predictors.

$$
\begin{aligned}
BMI \; &= \alpha + \beta_{AgeGroup} \cdot \mathbb{1}_{AgeGroup}(x) + \beta_{Sex} \cdot \mathbb{1}_{Sex}(male) \\
&\quad + \beta_{Employment} \cdot \mathbb{1}_{Employment}(x) + \beta_{Veg} \cdot \mathbb{1}_{Veg}(Yes) \\
&\quad + \beta_{Fruit} \cdot \mathbb{1}_{Fruit}(Yes) + \beta_{Year} \cdot \mathbb{1}_{Year}(x)
\end{aligned}
$$

Table 2: Estimates of the regression model coefficient

	BMI	
Predictors	Estimates	p
(Intercept)	-17.90	0.713
AgeGroup25-34	1.18	**<0.001**
AgeGroup35-44	2.50	**<0.001**
AgeGroup45-54	2.77	**<0.001**
AgeGroup55-64	3.28	**<0.001**
AgeGroup65-74	3.19	**<0.001**
AgeGroup [75+]	2.33	**<0.001**
Sex [Male]	0.05	0.411
Employment [In full-time education]	-0.87	**0.016**
Employment [In paid employment, self-employed or on gov't training]	0.67	**0.034**
Employment [Looking after home/family]	0.50	0.138
Employment [Looking for/intending to look for paid work]	0.23	0.525
Employment [Perm unable to work]	1.18	**0.001**
Employment [Retired]	0.66	**0.045**
Veg [Yes]	-0.42	**<0.001**
Fruit [Yes]	0.02	0.771
Year	0.02	0.377
Observations	25224	
R^2 / R^2 adjusted	0.048 / 0.047	

The estimated coefficients from this model are presented in Table 2, from which the reduced model can be determined by removing variables whose coefficients were not statistically significant at the 5% significance level. Sex, fruit intake, and year were all found to be non-significant and were therefore excluded. The removal of year indicates that there is no statistically significant evidence that BMI in Scotland changed over the years 2008–2012, after accounting for the remaining covariates.

3.2 Reduced Model

Table 3: Estimates of the reduced regression model coefficients

	BMI	
Predictors	Estimates	p
(Intercept)	25.14	**<0.001**
AgeGroup25-34	1.18	**<0.001**
AgeGroup35-44	2.51	**<0.001**
AgeGroup45-54	2.78	**<0.001**
AgeGroup55-64	3.29	**<0.001**
AgeGroup65-74	3.21	**<0.001**
AgeGroup [75+]	2.35	**<0.001**
Employment [In full-time education]	-0.87	**0.016**
Employment [In paid employment, self-employed or on gov't training]	0.66	**0.035**
Employment [Looking after home/family]	0.47	0.162
Employment [Looking for/intending to look for paid work]	0.24	0.511
Employment [Perm unable to work]	1.18	**0.001**
Employment [Retired]	0.65	**0.049**
Veg [Yes]	-0.42	**<0.001**
Observations	25224	
R^2 / R^2 adjusted	0.048 / 0.047	

$$
\begin{aligned}
BMI \; &= \alpha + \beta_{Age[\text{25-34}]} \cdot \mathbb{1}_{Age}([\text{25-34}]) + \beta_{Age[\text{35-44}]} \cdot \mathbb{1}_{Age}([\text{35-44}]) \\
&+ \beta_{Age[\text{45-54}]} \cdot \mathbb{1}_{Age}([\text{45-54}]) + \beta_{Age[\text{55-64}]} \cdot \mathbb{1}_{Age}([\text{55-64}]) \\
&+ \beta_{Age[\text{65-74}]} \cdot \mathbb{1}_{Age}([\text{65-74}]) + \beta_{Age[\text{75+}]} \cdot \mathbb{1}_{Age}([\text{75+}]) \\
&+ \beta_{Emp[\text{FT Edu}]} \cdot \mathbb{1}_{Emp}([\text{FT Edu}]) + \beta_{Emp[\text{Employed}]} \cdot \mathbb{1}_{Emp}([\text{Employed}]) \\
&+ \beta_{Emp[\text{Unable}]} \cdot \mathbb{1}_{Emp}([\text{Unable}]) + \beta_{Emp[\text{Retired}]} \cdot \mathbb{1}_{Emp}([\text{Retired}]) \\
&+ \beta_{Veg} \cdot \mathbb{1}_{Veg}(Yes)
\end{aligned}
$$

With this reduced model, the fitted regression model is:

$$
\begin{aligned}
BMI \; &= -17.9 + 1.18 \cdot \mathbb{1}_{Age}([\text{25-34}]) + 2.5 \cdot \mathbb{1}_{Age}([\text{35-44}]) + 2.77 \cdot \mathbb{1}_{Age}([\text{45-54}]) \\
&+ 3.28 \cdot \mathbb{1}_{Age}([\text{55-64}]) + 3.19 \cdot \mathbb{1}_{Age}([\text{65-74}]) + 2.33 \cdot \mathbb{1}_{Age}([\text{75+}]) \\
&- 0.87 \cdot \mathbb{1}_{Emp}([\text{FT Edu}]) + 0.67 \cdot \mathbb{1}_{Emp}([\text{Employed}]) \\
&+ 1.18 \cdot \mathbb{1}_{Emp}([\text{Unable to work}]) + 0.66 \cdot \mathbb{1}_{Emp}([\text{Retired}]) \\
&- 0.42 \cdot \mathbb{1}_{Veg}(Yes)
\end{aligned}
$$

The baseline category for this model is individuals aged 16-24, not in full-time education and not meeting the recommended daily vegetable intake. Interpreting the coefficients relative to this baseline: BMI increases substantially with age, peaking in the 55-64 age group with a coefficient of 3.28, before declining slightly in the older age groups. Individuals unable to work have the highest BMI relative to the baseline, with a coefficient of 1.18, while those in full-time education have a lower BMI than the baseline, with a coefficient of -0.87. Employed and retired individuals show small positive associations with BMI of 0.67 and 0.66 respectively. Finally, individuals who meet the recommended daily vegetable intake have a BMI that is on average 0.42 units lower than those who do not.

3.3 Model Diagnostics

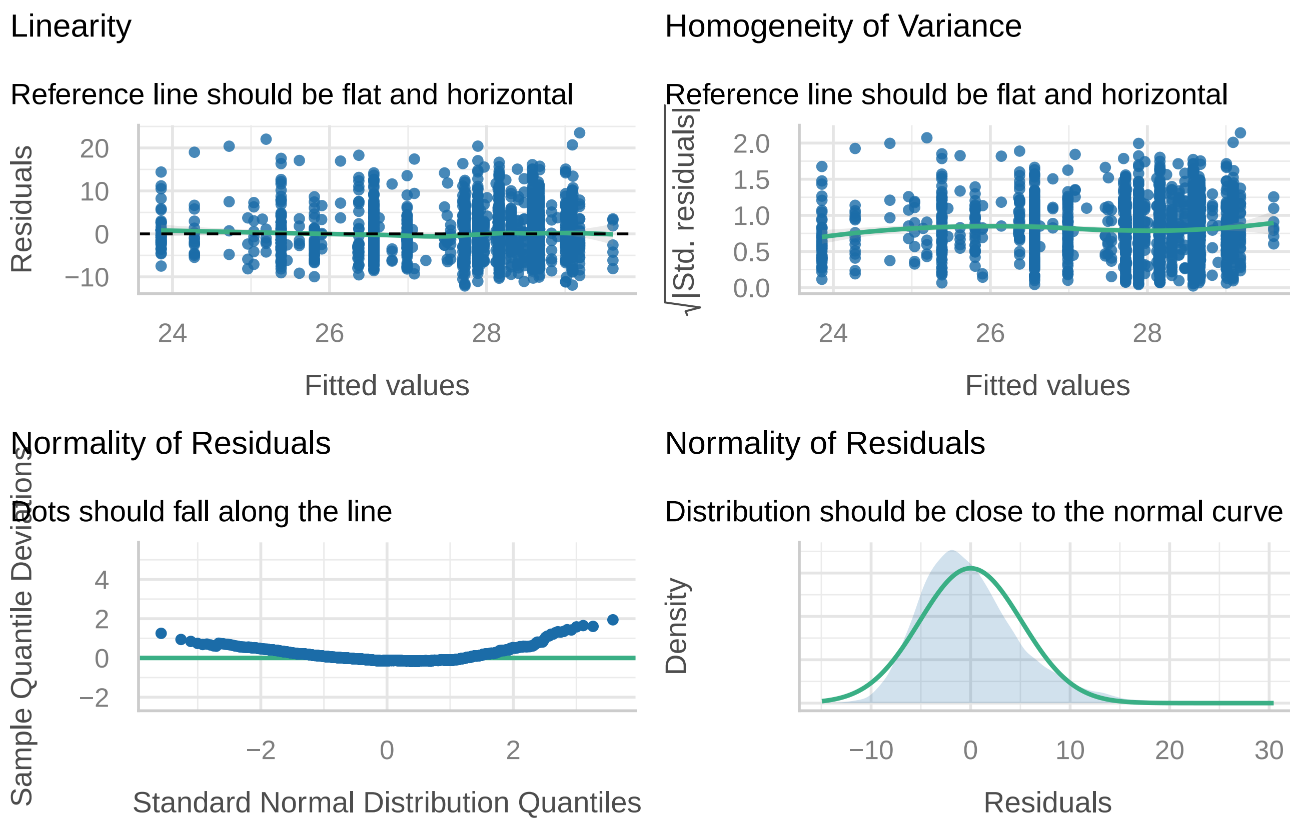

Figure 6: Final model diagnostic plots

The model assumptions were assessed using the diagnostic plots displayed in Figure 6. The linearity plot shows residuals scattered randomly around zero, suggesting the linearity assumption is reasonably satisfied. The homogeneity of variance plot similarly shows no strong pattern in the spread of residuals across fitted values, indicating constant variance is approximately met. However, the Q-Q plot and density plot indicate that the residuals are not normally distributed, displaying a left skew. This suggests some caution should be applied when interpreting the results of this model, though given the large sample size, the parameter estimates remain approximately valid by the Central Limit Theorem.

4 Conclusion

The aim of this investigation was to determine if BMI in Scotland changed between 2008 and 2012 and if BMI distribution differed by demographic, socio-economic, and lifestyle factors. The multiple regression analysis indicated that year was not a statistically significant predictor of BMI, suggesting that there is no evidence that the average BMI in Scotland changed over the period of 2008 to 2012.

Several demographic and socio-economic variables were found to be associated with BMI, with age group showing the strongest effect. BMI generally increased with age and peaked at the 55-64 age group before decreasing slightly in older groups. Employment status also showed significant association, with individuals who were unable to work having a higher BMI on average, and those in full-time education had a lower BMI relative to the baseline group. Meeting the recommended vegetable intake was associated with a small reduction in BMI. Sex and fruit intake were not found to be statistically significant predictors.

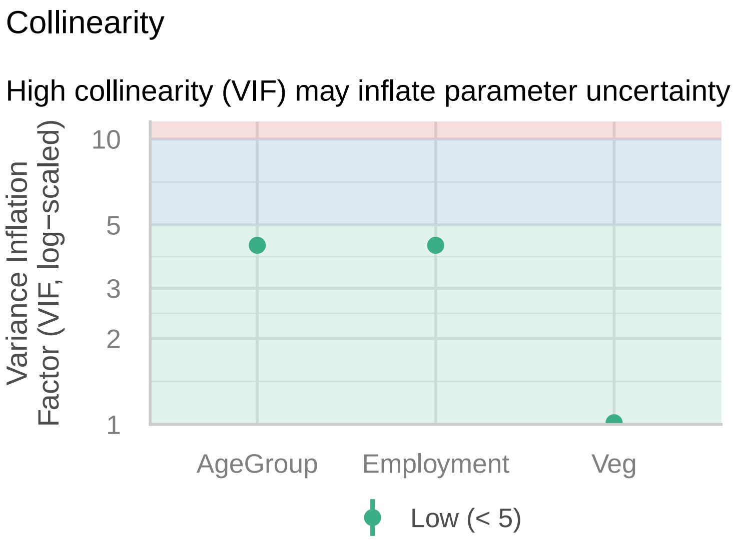

Figure 7: Diagnostics for multicolinearity

Diagnostics indicated that all the necessary assumptions were met other than normality of residuals. But due to the large sample size, the regression estimates could be considered approximately valid, although this assumption violation should be considered when interpreting the results.

Future work on this topic could involve investigating interaction effects between variables, and analysing additional lifestyle and socio-economic variables such as smoking, drinking, physical activity, income level, etc., to further investigate factors impacting BMI. Extending the analysis to include more recent years if the Scottish Health Survey could also allow a more in-depth investigation into how BMI in Scotland has changed over the years.